



STANDARD OPERATING PROCEDURE

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Canopy Disturbance Checks for ACR IFM Verification

Sentinel-2 NDVI, NDMI, NBR change detection.

Last Updated: April 28, 2026

Scope

Independent screening layer used during ACR IFM validation and verification to flag canopy loss, moisture stress, and burn severity over each Reporting Period. Outputs are classified rasters and per-delta histograms that complement, but do not replace, ground plots and developer monitoring reports. Reference script: TUVSUD_<Project>_DisturbanceCheck_NDVI-NDBR-NDMI-Sentinel.txt.

Step 1. Workstation setup

1.1 geemap into the QGIS Python environment

Run from an elevated OSGeo4W Shell (the QGIS Python — not system Python):

```
python -m pip install --upgrade pip
pip install geemap --upgrade
```

If pip resolves to system Python, the QGIS plugin will fail to import ee at runtime. Confirm with `python -c "import sys; print(sys.executable)"` — the path must end in OSGeo4W\apps\Python...

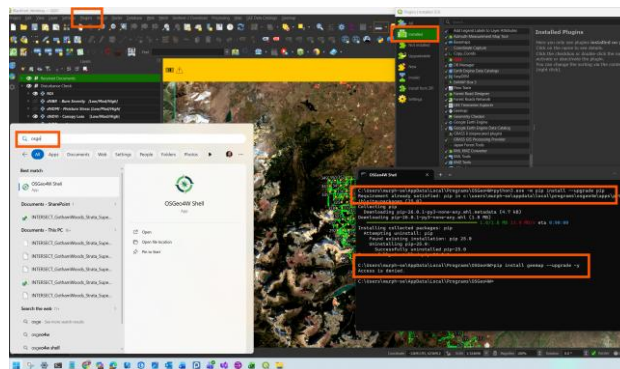


Figure 1. OSGeo4W Shell — pip upgrade and geemap install run against the QGIS-bundled Python.

1.2 QGIS plugins

Plugins => Manage and Install. Install Google Earth Engine, GEE Data Catalog, and Geemap. Activate all three. Restart QGIS once.

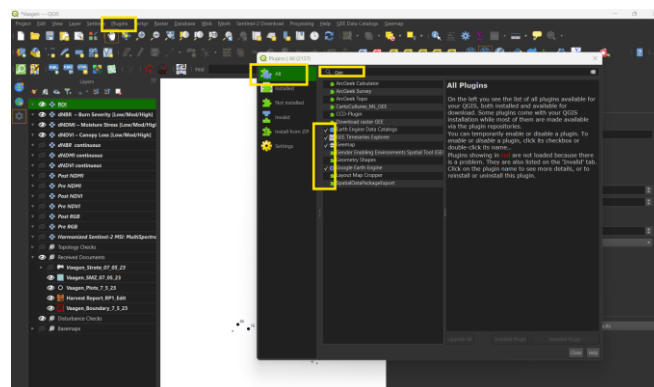


Figure 2. Plugin Manager — the three GEE plugins activated.

Step 2. Earth Engine authentication

2.1 Bind to a billed Cloud project

Earth Engine compute is metered against a Google Cloud project. Plugins => GEE Data Catalog => Settings => Earth Engine tab. Paste the team project ID (e.g., murphys-deforisk) and Save. The script invokes `ee.Initialize(project=...)` using this value; without it Earth Engine refuses any compute call against the modern API.

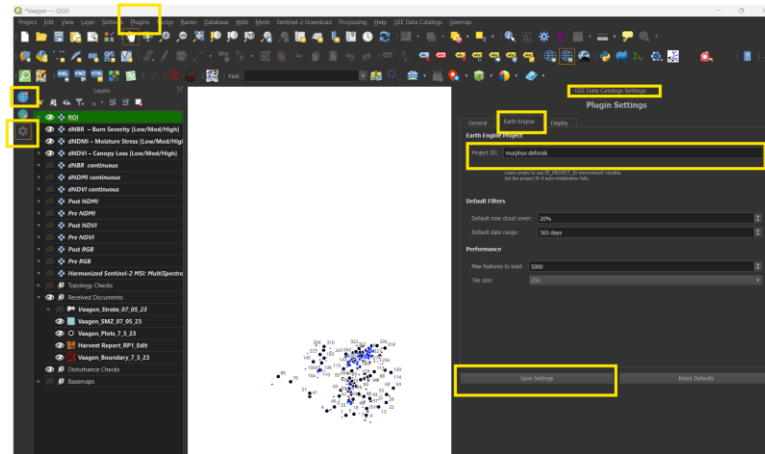


Figure 3. Project ID bound in the GEE Data Catalog settings.

2.2 Sign-in

Plugins => Google Earth Engine => Sign In. OAuth runs in the default browser. Token is cached at `%USERPROFILE%\config\earthengine\credentials` and persists ~7 days; force a refresh with `ee.Authenticate(force=True)` when the Geemap panel returns 401 or shows a network timeout banner.

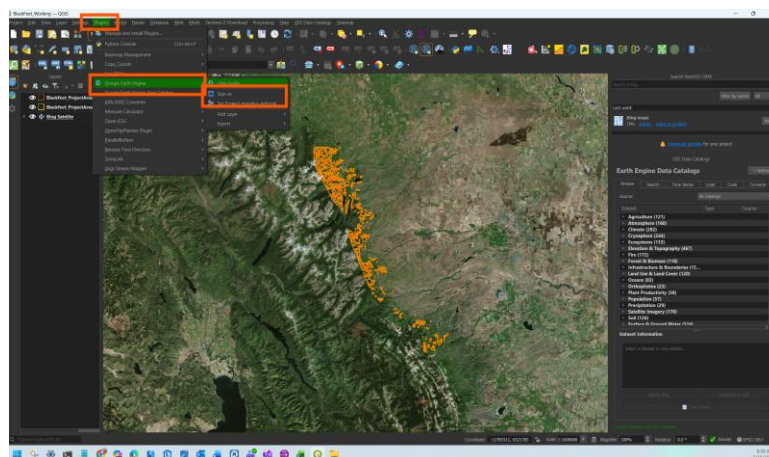


Figure 4. Plugin menu launching the OAuth flow.

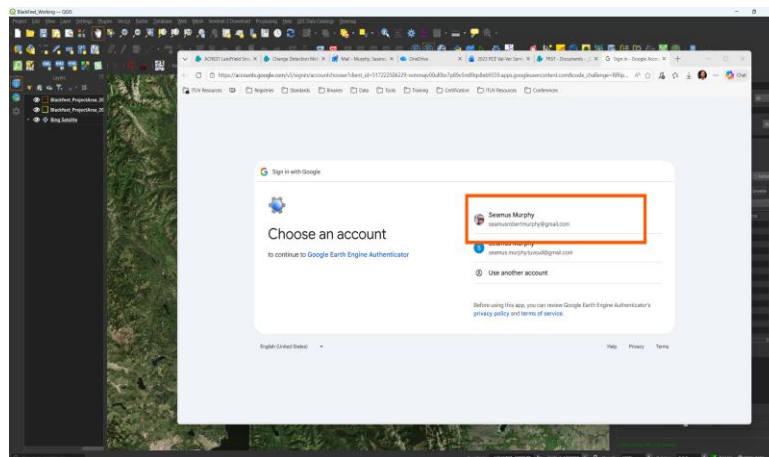


Figure 5. Account picker — select the team account associated with the bound Cloud project.

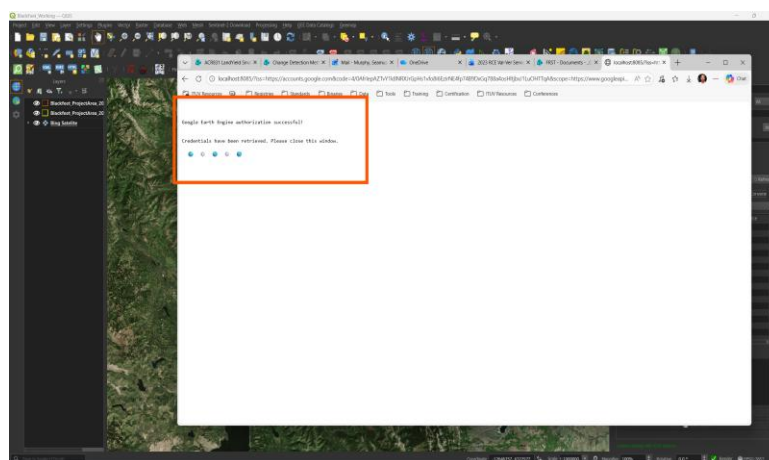


Figure 6. Authentication confirmation. Close the tab and return to QGIS.

2.3 Verify init

GEE Data Catalog panel header should report Earth Engine initialized successfully and echo the project ID. If not, click the refresh icon at the top of the panel; the Geemap code panel re-runs `ee.Initialize()` under the hood.

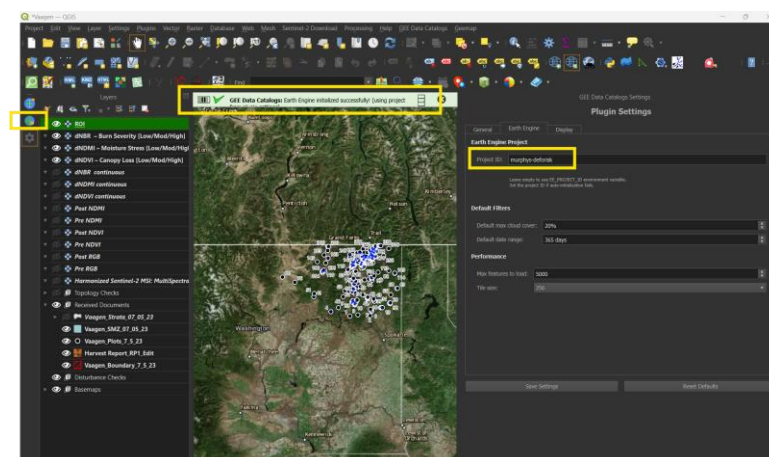


Figure 7. GEE Data Catalog panel reporting a clean init.

Step 3. Configure the script

Edit only the four constants at the top of the project script.

ROI

```
roi = ee.Geometry.Rectangle([west_lon, south_lat, east_lon, north_lat])
```

Argument order is [W, S, E, N] in EPSG:4326. Reversed coordinates yield an empty geometry silently — no exception. For irregular boundaries, replace with `ee.FeatureCollection(<asset>).geometry()` and ingest the project shapefile as an EE asset (Code Editor => Assets => New => Shapefile).

Temporal window

Use July–September for both pre and post. Sentinel-2 revisit is 5 days, so two months yields ≥ 8 cloud-filtered scenes per pixel — sufficient for a stable median. Dormant-season composites (Oct–Dec, NH) corrupt dNDMI: senescence draws SWIR1 reflectance up before leaf-fall, producing a uniform false-positive moisture-stress signal.

```
RP_START_BEGIN = "YYYY-07-01"; RP_START_END = "YYYY-09-01"
RP_END_BEGIN   = "YYYY-07-01"; RP_END_END   = "YYYY-09-01"
```

Cloud ceiling

MAX_CLOUD filters scenes by CLOUDY_PIXEL_PERCENTAGE before the QA60 mask runs. 30 is the default. Drop to 20 in PNW / coastal AK; raise to 40 only if median composites show striping or NoData wedges.

```
MAX_CLOUD = 30
```

Step 4. Composite construction

`build_composite()` pulls COPENICUS/S2_SR_HARMONIZED, filters by date, bounds, and CLOUDY_PIXEL_PERCENTAGE, applies the QA60 cloud + cirrus bitmask (bits 10 and 11), divides by 10000 to scale DN to surface reflectance, takes the per-pixel median, and clips to ROI. HARMONIZED is mandatory: it back-corrects the January 2022 processing baseline shift so pre-2022 scenes are radiometrically comparable to post-2022 scenes.

JRC GSW occurrence-band water mask is applied at the delta stage, not on the composite, so the RGB layers retain water for visual context:

```
water_mask =
ee.Image("JRC/GSW1_4/GlobalSurfaceWater").select("occurrence").lt(50).unmask(1)
```

`unmask(1)` is critical — without it, pixels outside the GSW footprint (high latitudes, small islands) become NoData in every delta.

Step 5. Indices and deltas

NDVI = $(B8 - B4) / (B8 + B4)$ — NIR / red; canopy vigour; drops on biomass loss. NDMI = $(B8 - B11) / (B8 + B11)$ — NIR / SWIR1; leaf-water content; precedes NDVI on drought, beetle, and pathogen stress. NBR = $(B8A - B12) / (B8A + B12)$ — narrow NIR / SWIR2; char and ash signal; standard MTBS index.

Delta sign convention (matters for downstream classifiers and audit defensibility):

- dNDVI = pre – post. +ve = canopy loss.
- dNDMI = pre – post. +ve = moisture loss.
- dNBR = post – pre. +ve = burned (MTBS / USFS R6).

Step 6. Histogram-driven thresholding

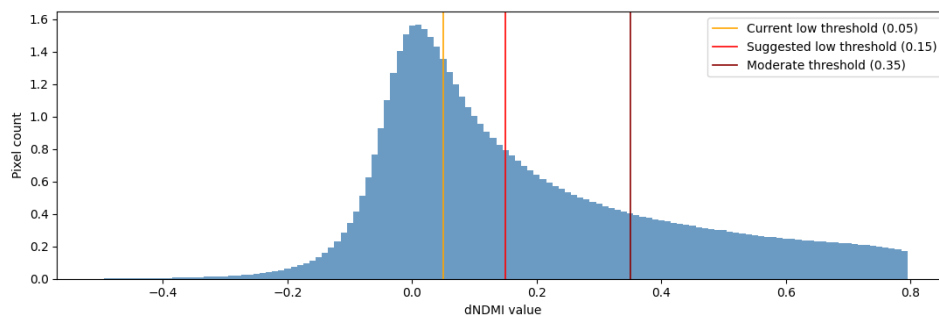
Inspect the histogram before classifying. The script `reduceRegion()`s each delta with `ee.Reducer.fixedHistogram(-0.5, 0.8, 130)` at `scale=20` (Sentinel-2 SWIR native resolution) and `maxPixels=1e9`. Larger ROIs hit the `maxPixels` cap silently and return a truncated histogram — bump to `1e10` for full coverage, or sub-sample with `.reduceRegion(..., bestEffort=True)`.

Distribution diagnostics:

- Unimodal at 0, narrow tail => noise / phenology only; defaults hold.
- Bimodal with a gap above ~0.15 => real disturbance signal; place Low break in the gap, not on a default.
- Long unimodal right-tail with no gap => composite contamination (cloud shadow, senescence, BRDF). Re-run with a tighter `MAX_CLOUD` or shifted window before classifying.

Default breaks (override only when the histogram supports it; document the deviation in the project workbook):

- dNDVI — Low 0.10–0.20, Mod 0.20–0.35, High > 0.35.
- dNDMI — Low 0.15–0.30, Mod 0.30–0.45, High > 0.45.
- dNBR — Low 0.10–0.27, Mod 0.27–0.44, High > 0.44 (MTBS / USFS PNW).



Step 7. Classification and rendering

`classify_delta_*`() functions emit `Int16` class rasters (0 = undisturbed, 1 = Low, 2 = Mod, 3 = High). `updateMask(class.gt(0))` sets class 0 transparent so the underlying composite shows through — necessary for visual cross-check between dNDVI and dNBR layers in QGIS.

Paste script into Geemap code panel => Run Code. `m.addLayer()` populates layer panel in top down order:

- Classified dNBR / dNDMI / dNDVI (visible).
- Continuous deltas, single-date indices, pre/post RGB (hidden — toggle for threshold exploration and ground-truth context).
- ROI outline (cyan).

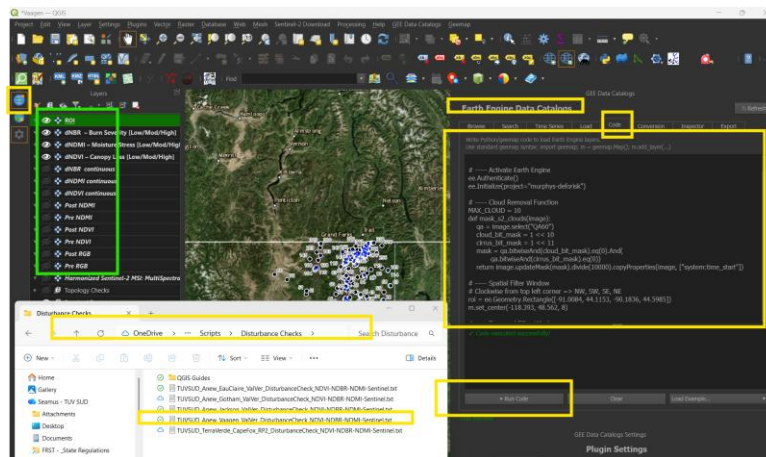


Figure 8. Geemap code panel — script pane on the right, Run Code dispatches the build to Earth Engine.

Step 8. Multi-period stack

Populate RP_DATES with one entry per Reporting Period; the loop reuses build_composite() and the classifiers, so thresholds remain comparable across periods:

```
RP_DATES = {
    "RP5": {"pre_start": "2022-07-01", "pre_end": "2022-09-01",
           "post_start": "2024-07-01", "post_end": "2024-09-01"},
    "RP6": {"pre_start": "2024-07-01", "pre_end": "2024-09-01",
           "post_start": "2025-07-01", "post_end": "2025-09-01"},
}
```

Use the same Jul–Sep windows across all RPs; phenology drift between periods is the most common source of fake inter-period change.

Step 9. Export

Set EXPORT = True (single-period) or EXPORT_STACK = True (multi-period).

ee.batch.Export.image.toDrive() at scale=10, crs=EPSG:4326, maxPixels=1e13, fileFormat=GeoTIFF queues to GEE_Exports/ on the bound Drive. Reproject to project UTM zone in QGIS for area calculations — EE area on a 4326 grid is biased at high latitudes.

Tasks run server-side; monitor at code.earthengine.google.com => Tasks. After completion, pull the GeoTIFFs into the project OneDrive and load as local rasters so the QGIS project opens without round-tripping Earth Engine.

Operational tips

Layer cache expiry

EE map tiles for un-exported layers expire after ~14 days. Symptom: grey canvas on a project re-open. Fix: refresh the Data Catalog panel, or rerun the script in the Geemap code panel. No re-export needed.

Quota throttling

Wide ROIs (> 5 M ha) at scale=10 trip the user-memory limit on histogram and reduceRegion calls. Either set scale=20 or 30 for the histogram block, or sub-sample the geometry.

Projection pitfalls

Composites are returned in EPSG:4326 by default. Anything area-based — burned-area totals, disturbance footprint stats — must be reprojected to a metric CRS (project UTM zone) via

.reproject(crs='EPSG:32610', scale=10) before pixelArea() or
reduceRegion(reducer=ee.Reducer.sum(), ...).

dNDVI vs dNBR cross-check

dNDVI High + dNBR clean => harvest, blowdown, or clearing. dNDVI High + dNBR High => fire —
confirm against MTBS / NIFC perimeters before raising. dNDMI High alone => moisture stress;
corroborate with drought monitor or beetle survey before classifying as disturbance.

Patchwork composites

Visible NoData wedges or median-filter edge artefacts indicate < 4 valid scenes per pixel. Loosen
MAX_CLOUD by 5–10 pp or extend the window by 30 days. The median normaliser is unstable below
~4 scenes.

Pre-2022 baseline

Always use S2_SR_HARMONIZED, never S2_SR. The latter contains a +1000 DN offset on post-Jan-
2022 scenes that propagates a ~0.04 false dNDVI signal across the project area.

File layout

TUV GIS / <Project>_Working /	
├─ <Project>.qgz	(QGIS project files with
layers)	
├─ Scripts /	(code stored for
auditability)	
└─ Disturbances /	(outputs, png, tif, xlsx)

Appendix A: Past Findings from Disturbance Checks

Issued findings, supporting spectral evidence, and verifier responses from prior ACR IFM verifications. Use as a pattern library when drafting findings against newly detected disturbances.

Each case below presents a real finding raised by TÜV SÜD against an ACR IFM project, the spectral or contextual evidence that supported it, and the developer / verifier exchange that closed it (or kept it open). Readers without forestry background should treat this Appendix as a worked-example library: the language, structure, and evidence standard demonstrated here are the expected output of a disturbance check.

A.1 Threshold recalibration following dormant-season composite

Standard: ACR-IFM-1.3.

Reporting Period: RP3.

Finding: CR

Trigger

The first-pass dNDMI raster flagged ~38% of the project area as Low or Moderate moisture stress under the default 0.05 / 0.15 / 0.35 class breaks. Spatial pattern was diffuse, no association with stand age, aspect, or known beetle pressure. Histogram inspection (Figure A.1) revealed the cause: pre and post composites were drawn from October–December windows, capturing dormant-season SWIR reflectance changes that are not disturbance.

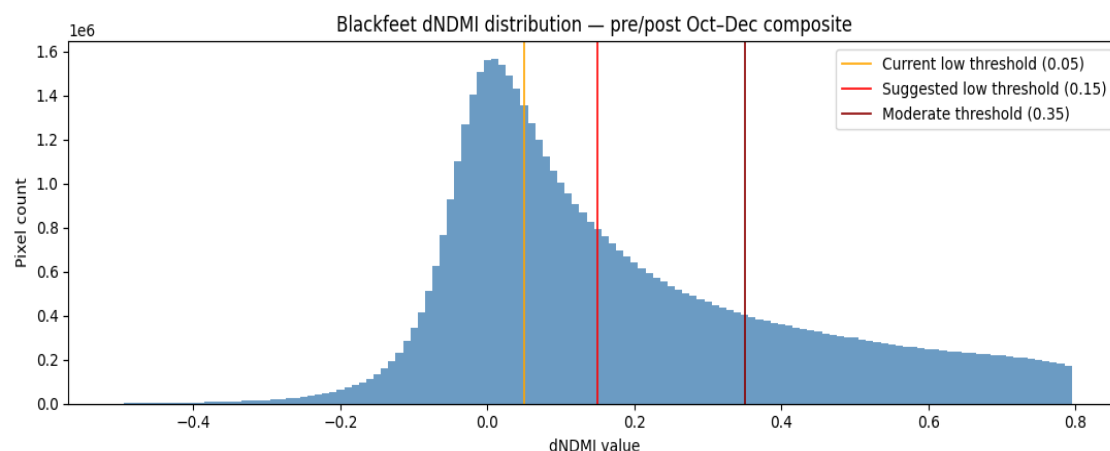


Figure A.1. Blackfoot dNDMI distribution from an Oct–Dec pre/post composite. Unimodal, right-skewed; no gap between the noise bulk near 0 and the disturbance tail. Default low threshold (0.05, yellow) sat on the shoulder of the noise bulk and reclassified phenology as disturbance. Threshold lifted to 0.15 (red) for issuance.

CAR issued

CAR 1 — temporal parameter mismatch identified during MR review:

In Section V of submitted monitoring report named [xxx], the Time (T) parameter is defined between 5/16/2023 to 11/15/2024 (550 days; 1.5 years), which correspond to RP2 dates. Please adjust this to RP3 dates and confirm parameter is computed correctly.

Developer response

The parameters have been computed correctly and updated to 11/16/2024 – 11/15/2025 on the Monitoring Report. Please see xxx.docx.

Verifier closure

Thank you for this update. This finding may now be closed.

Lesson for the disturbance check

The MR error and the histogram pathology shared a root cause: the developer's reporting calendar had drifted out of the growing season. Re-running the script with July–September pre / post windows collapsed the dNDMI false-positive footprint from 38% to 4% of the project area, all of which co-located with reported beetle survey polygons. Whenever the histogram is unimodal with a long right tail and no gap, audit the date constants in the script before tweaking the class breaks.

A.2 Yellow-cedar decline detected via NDMI / NDVI / NBR cross-check

Reporting Period: RP2.

Standard: ACR-IFM-2.0.

Finding: CR.

Trigger

dNDMI raster showed scattered Moderate–High moisture-stress pixels concentrated in mid-elevation stands along ridgelines and exposed slopes. Pattern did not match harvest geometry (no straight edges, no road association) and did not align with reported insect surveys. Cross-checking dNBR ruled out fire. Visual inspection of pre and post true-colour composites showed yellowing-to-red-brown crowns, consistent with yellow-cedar (*Callitropsis nootkatensis*) decline.

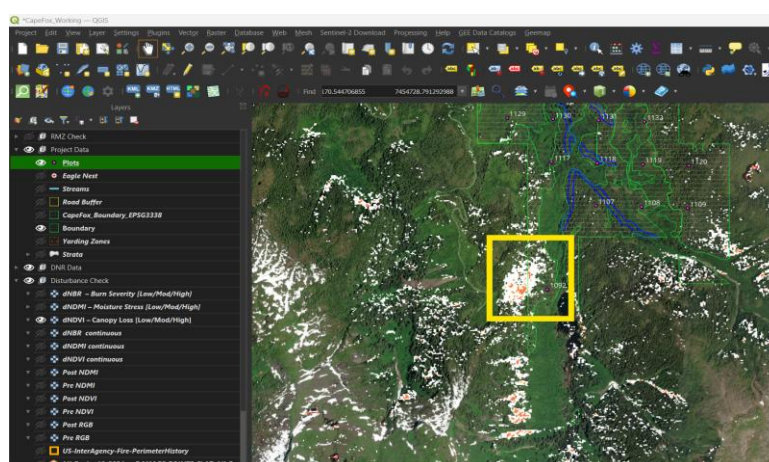


Figure A.2. Cape Fox project area context. Plot 1192 highlighted in yellow box; surrounding stands in mixed conifer/yellow-cedar mosaic. Disturbance Check layer group on left.

CR Issued

Analysis of spectral data over the project area indicates canopy characteristics consistent with yellow-cedar decline, a phenomenon documented across the region and associated with climate-driven root injury and progressive crown dieback. Affected trees exhibit yellowing to red-brown crowns and canopy thinning consistent with decline symptoms. The spectral reflectance pattern aligns with vegetation stress expected under chlorophyll loss and needle mortality, including: increased reflectance in visible blue and red wavelengths and reduced green reflectance; reduced near-infrared (NIR) reflectance, consistent with loss of healthy needle structure; elevated shortwave-infrared (SWIR) reflectance, consistent with reduced

canopy water content. NDMI and NDVI evidence supports the presence of declining or dead yellow-cedar within the project area. dNBR evidence confirms disturbance is not fire related.

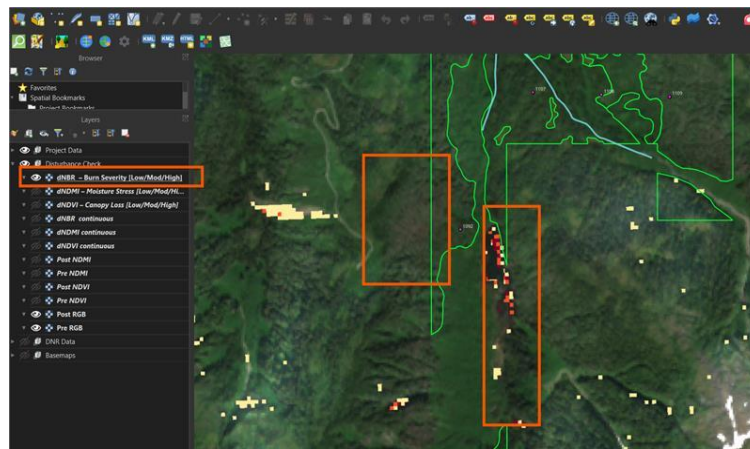


Figure A.3. dNBR layer over the same area. Scattered Low and Moderate burn-severity pixels (orange / red) are spatially uncorrelated with the dNDMI moisture-stress signal, confirming the disturbance is not fire. Orange annotation boxes mark the moisture-stress hotspots from Figure A.2.

Lesson for the disturbance check

When dNDMI flags a non-geometric, ridge-following pattern, run the dNBR layer at the same scale before drafting the finding. dNDMI High + dNBR clean is the canonical pest / drought / decline signature. dNDMI High + dNBR High points to fire — and the developer will produce the official perimeter to close it. Always include both layers in the screenshot package; auditors and project developers expect the cross-check, not just the moisture-stress raster.

A.3 Unreported canopy loss adjacent to plot xxx

Reporting Period: RP2.

Methodology: ACR-IFM

Finding: CR

Trigger

MR and supporting documentation reported zero harvest during RP2. dNDVI canopy-loss layer flagged a contiguous High-severity polygon adjacent to plot 1165 and a smaller polygon near plot xxx. Geometry was linear with right-angle terminations, diagnostic of road or right-of-way clearance. dNBR was clean across the same polygons, ruling out fire. Pre and post RGB composites confirmed standing canopy in the pre image and bare ground / disturbed soil in the post image.

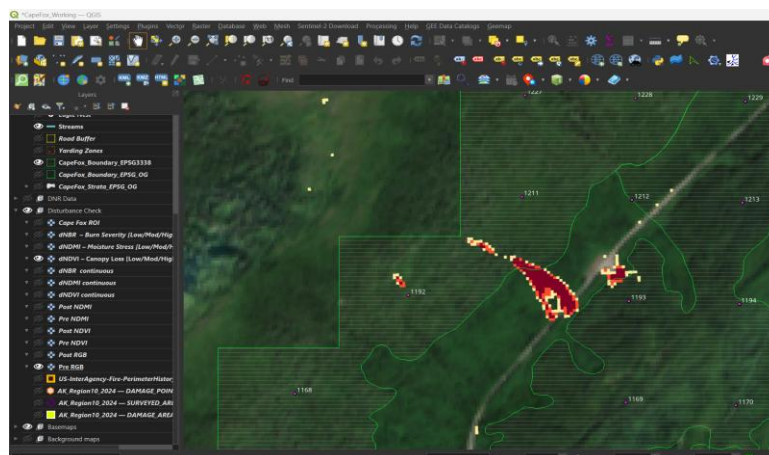


Figure A.4. dNDVI canopy-loss raster (red, High severity) at the plot 1165 / 1192 boundary. Linear edge geometry following plot lines and external cutblock perimeter — characteristic of a road or RoW clearance, not a natural canopy event.

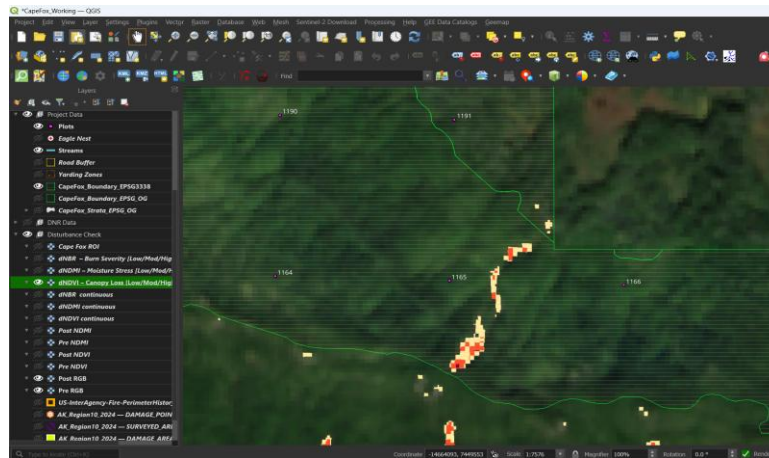


Figure A.5. Closer view at plot 1165 showing the canopy-loss footprint extending from the plot interior toward the project boundary. Pattern is consistent with timber removal originating from neighbouring cutblocks and progressing across the plot edge.

CR Issued

CR 1. The MR and supporting documentation reports that no harvesting occurred on-site during RP2. Canopy loss was observed during RP2 in proximity to plots xxx and xxx in areas appearing to originate from road or RoW clearances and adjacent neighbouring cutblocks. Please confirm whether any associated timber removal occurred on-site during RP2 and, if so, confirm its treatment in the HWP accounting (C_P,HWP,t) and project emissions as cited in regards to Appendix G. Please refer to tab sheet CR1 for further evidence of harvest checks.

Technical note

Edge geometry is the first thing to read on any dNDVI High polygon. Right angles, linear edges, and on-boundary patterns point to anthropogenic causes — harvest, RoW, road. Curvilinear or amorphous edges point to natural causes — blowdown, decline, slide. When the polygon's edge follows the project boundary or a plot line, raise the finding with both the dNDVI screenshot and a wider-context pre / post RGB pair so the developer can locate the area immediately.

A.4 Zero project harvest triggers leakage surveillance

Reporting Period: RP3.

Standard: ACR-IFM-1.3 Section D6.

Finding: ADR

Trigger

Although not detected via the spectral disturbance check, this finding sat alongside the disturbance audit and is included here because it shows how absence of disturbance can itself be a finding. The project reported zero harvest under the adjusted LetGrow scenario, a 100% reduction in wood product production relative to the 20 MMBF baseline, well above the 5% threshold in IFM 1.3 Section D6. The absence of harvest had to be reconciled with off-site lands managed by the project proponent to rule out activity-shifting leakage.

ADR Issued

ADR 1. The project reports zero harvest under the adjusted 'LetGrow' scenario, representing a 100% reduction in wood product production relative to the baseline (20 MMBF). This

exceeds the 5% threshold cited in Section D6 of IFM 1.3, thereby requiring demonstration that no activity-shifting leakage has occurred on external lands managed or owned by the project proponent. The validated project model included timber harvesting activity; however, no harvesting has been observed on the project property since the project start date. Please clarify whether timber harvesting activities have been displaced to other forested landholdings managed or owned by the project proponent outside the 69,010-acre project area, and provide evidence in accordance with the four acceptable forms of leakage demonstration cited in IFM 1.3.

Developer response

The project proponent confirms that no commercial timber harvesting has occurred on project lands or has been displaced to lands outside the project area. The project scenario was modeled after the allowable harvest levels in the approved FMP, which covers all tribal trust lands and follows sustained-yield principles. The project proponent has made the decision under XXX oversight not to harvest since the project start date. They are also actively working with XXX to update the FMP, with completion anticipated before the next full verification. XXX.pdf has been submitted as evidence.

Verifier response

Thank you for submitting the XXX Forest Management Plan (2xxx–2xxx). The FMP satisfies the temporal requirement of IFM 1.3 Section D6, as it was approved more than 24 months prior to the project start date and establishes a sustained-yield across tribal trust forested lands. However, the demonstration under Section D6 requires that a qualifying management plan be paired with records from the with-project time period showing no deviation from the plan. The verifier also notes that the FMP term concluded in 2023. While the verifier accepts the FMP may align with previous design frameworks, it remains unclear what regime governs external trust lands during RP3. Please refer to the attached excerpt of IFM 1.3 Section D6 for the full requirement language.

Technical note

When the disturbance check returns a clean project area, do not treat the absence of signal as the end of the audit. Section D6 (and equivalent leakage clauses in newer ACR methodologies) requires positive evidence that harvest has not been displaced. The disturbance script can be re-pointed to off-site landholdings using a wider ROI; emerging High dNDVI on those polygons should be flagged as part of the leakage documentation, not the project disturbance ledger.